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An Extreme Learning Based Forest Fire Detection Using Satellite Images with Remote Sensing Norms

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Abstract- When it comes to maintaining ecological and societal harmony, forests play a crucial role in the ecosystem. For a variety of reasons, forest fires pose the greatest threat to forests of this significance. The advent of satellite technology has made it possible to continuously monitor and suppress forest fires, which pose a significant hazard to humans and other living animals. When smoke is visible in the sky, it means that wildfires are burning in the forest. To prevent damages and other fire disasters with societal repercussions, fire detection is an essential component of fire alarm systems. It is crucial to predict the formation and behaviour characteristics of flames in order to battle forest fires, and accurate fire detection from visual situations is critical for avoiding large-scale fires. It is much easier to recognize the regions impacted by fires and their intensity using satellite photos acquired with improving technologies for this purpose. Learning based Remote Fire Detection (LBRFD) is a new deep learning method that has been developed to enhance fire detection accuracy. To test how well LBRFD works, it is cross-validated with the usual Convolutional Neural Network (CNN) model of learning. To train the satellite images to distinguish between fire and non-fire images, a new framework is created to define the framework's efficiency. Then, the region where the fire occurred in the satellite image is extracted using a local binary pattern, which decreases the number of false positives and this technique is called LBRFD.

Index Terms—Extreme Learning, Forest Fire Detection, Satellite Images, Remote Sensing, LBRFD, Convolutional Neural Network, CNN.

I. INTRODUCTION

Among the most destructive environmental catastrophes, forest fires kill many people and destroy countless homes. It has had global, permanent impacts in the last many years. There is an ever-present risk to natural systems, infrastructure, and human lives posed by forest fires [1]. There have been several forest and wildfires in the past. Landscape design, structure, and, in the long run, the organism diversity in environments are all profoundly impacted by fires. Considerations such as biological variety, soil nutrient availability, and plant community growth shape forest fires' ecological significance. Even though they pose a threat to human life, fires are a major environmental problem due to the extensive economic and ecological harm they do. Humans or mechanical equipment are used to keep an eye on the environment in traditional fire defense systems [2]. Particle collection, thermal collection, and airflow accountability measurement are all considered the mainstays of fire smoke identification methods. Trees, fallen leaves, lumber and other materials cover a large area in the forest. These compounds contribute to the initial growth of the fire. The following figure, Fig-1 represents the dataset sample images of the proposed approach.

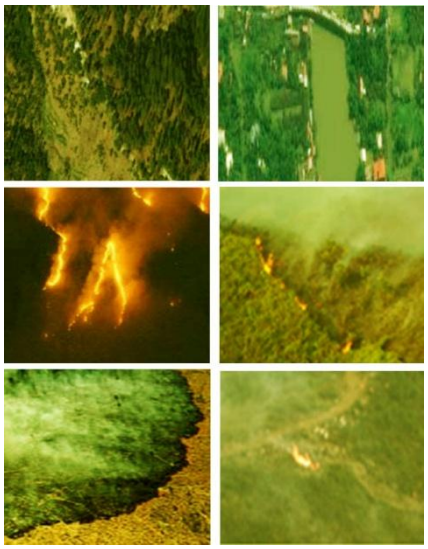

Fig.1 Dataset Samples

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A fire might start for a variety of reasons, such as smoking, excessive heat in the summer, or an event that included fireworks [3]. As soon as it begins to burn, a fire will continue to burn until it dies out completely. There will be less damage and less money spent on putting out the fire if it is found quickly. For that reason, fire detection is of the utmost importance here. Quickly notifying fire personnel and locating the exact location of the incident can have a positive impact. From optical cameras and sensors to tower and satellite monitoring, the government employs a wide range of methods for detecting fires. There are a plethora of other approaches of putting out fires. One of the most common is using floating water reservoirs to put out fires, which is common in many places, or lighting flame to warm up areas [4][5]. In countries in the Middle East, these elements carry it away, and then it is burnt in a designated place without fuel. But down under, they just light these areas on fire and let them burn out on their own, keeping the local animals and humans safe. Until the particles land on the sensors and trigger them, no alert will go off. Forest fire managers and the government have a significant problem in detecting wildfires early on, which is crucial for protecting environmental spaces [6][7]. The most significant factor in the reduction of forest acreage is forest fires. In addition to reducing the need for human processes, this fire detection method aids in monitoring and protecting those places that are particularly difficult to safeguard. Regardless of the weather or time of day, the new method makes it easier to install devices that enable efficient monitoring in specific locations.

Additionally, satellite photos provided a means of fire monitoring, control, and damage assessment in order to comprehend an advantageous fire range in relation to burn regions [8]. As with the initial fire materials, the uniformity of hue is the primary criterion for categorizing this fire. We ran the numbers on the system's differential matrix, detection matrix, threshold value, and forest fire analysis using the settings we suggested. Human activities have put the woods at serious risk across the board. Areas that are difficult to reach, lack the tools needed, and are understaffed. In addition, keeping tabs on these woods all the time is a huge pain and almost impossible [9]. Automated fire detection systems may be classified into three types, as shown in this study: ground, aerial, and carried. In order to distinguish between fire smoke and background smoke, the ground-based sensors employ a network of fixed black-and-white video cameras. Its main benefit is the high level of both temporal and geographic precision it offers. When it comes to early fire detection, though, these technologies still have their limitations. That is why a mechanism to detect fires quickly must be put into place [10].

II. Related Study

The increased frequency of forest fires in recent decades is mostly attributable to human-caused factors such as climate change and careless land management [11]. In recent years, forest fires have become more common as a result of the intensification of the climatic "El Niño" phenomenon, which is characterized by high temperatures and extended droughts. It takes too much time to notify the relief corps and needs well-organized logistics, but the most common way of detecting forest fires is visually, either from the ground or by utilizing a helicopter. It is clear from the recent occurrences (the most recent fires) that there are not enough safeguards in place to address the absence of early detection methods. For economic development cards like a Raspberry Pi, this article's goal is to compare and contrast the efficacy of several convolutional neural network (CNN) models that have already been trained to classify photographs of forest fires [11].

The dangers of forest fires are immense. It is really challenging to put them out after they have become catastrophes. A method for detecting forest fires using images taken by unmanned aerial vehicles (UAVs) is suggested in this article [12]. The first step in detecting smoke and making a preliminary forest fire discriminate is to apply the local binary pattern (LBP) feature extraction and support vector machine (SVM) classifier. The convolutional neural network (CNN) claims that it can reduce the amount of parameters and improve training performance through local receptive domain, weight sharing, and pooling, which makes it ideal for early stage fire identification. In this study, we offer an alternative approach to using convolutional neural networks to identify forest fires. Before adding a picture to a CNN network, it undergoes a series of preprocessing steps, including histogram equalization and smooth low-pass filtering. Images of actual forest fires are used to confirm the efficacy of the suggested strategy [12].

Any kind of flame, no matter how big or little, represents an unsavory setting, circumstance, or moment. If a fire were to break out, it might happen anywhere. The smoke detectors, however, are the most popular fire detection tools now in use [13]. In such case, smoke detectors will only go off in the event of a major fire. So, a method to identify fires in their early stages is necessary. This article presents the design of an image-based fire alarm system that primarily utilizes a laptop and camera. A procedure for detecting fires using CNNs and the developed technique achieves a 92% success rate [13].

One of the primary causes of environmental damage is forest fires. A quicker and more precise detection system can help reduce the amount of damage that fires can do, especially in their early phases when they are hard to spot [14]. This study introduces a method for autonomous

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early fire detection that does not require human intervention or service and is based on a very reliable system. The paper's meat and potatoes are an object identification approach we built using a convolutional neural network to provide the suggested system its autonomous capabilities. Instead of using outdated methods like lookout towers and satellite-based monitoring, we employ a live video feed from an unmanned aerial vehicle (UAV) that patrols the dangerous region to provide a greater field of vision over the monitored area. By utilizing both the optical and thermal cameras on board the UAV, we are able to improve our firing probability forecasts. Our web-based platform, built on top of the Node-RED software platform, may alert relevant parties in real-time and display the collected data. This document also details the process flow for building the web-platform [14].

Research communities are currently seeking ways for early identification and control of forest fires due to the increased frequency and severity of these fires in recent years [15]. Early detection from a wide area is only one of the many benefits of remote sensing utilizing computer vision techniques; other benefits include the ability to pinpoint the exact position and intensity of a fire. Recent experiments have demonstrated the potential of integrating computer vision with aerial platforms like manned and unmanned aerial vehicles, particularly small-size and low-cost UAVs, to detect, geolocation, and gather information about fire characteristics in forest fires. This study contributes to the literature by presenting a new approach to forest fire detection utilizing a single artificial neural network and a local binary pattern of flame and smoke signatures in colour and multi-color space. The majority of the training and assessment photos used in this article were captured from airborne platforms, which can pose some challenges. These include photographs with very small flame pixels, images with complicated backdrops, images with occluded flame and smoke areas, and images where the smoke blends into the background. The suggested approach has preserved a processing speed of 19 frames per second while achieving F1 scores of 0.84 for flame and 0.90 for smoke. In terms of recognizing difficult flame and smoke areas with varying sizes, colours, textures, and opacity, it has proven to be more effective than YOLOv3, YVML, random forests, and Bayesian classifiers [15].

III. METHODOLOGY

Conventional responses, such as warmth and smoke alarms, are utilized by the fire detection systems that are now in application. The fact that smoke and heat sensor alarms only have a single module means that they are unable to continuously monitor all of the potential fire hotspots. This is a significant drawback of these alarms. Maintaining vigilance is the only method that can be guaranteed to prevent a fire from spreading. Even if smoke detectors are planted in every crack and crevice of a forest, it will not be sufficient to provide an effective output on a continuous basis. It is almost impossible to establish smoke detectors in every part of a forest that is susceptible to flames. In the event that you require a greater quantity of smoke detectors, the cost will increase by a factor of times. The alerts that are generated by the proposed system are reliable and extremely accurate, and they may be generated within minutes of a fire or disaster occurring. Due to the fact that the whole monitoring network is driven by just one component of applications, expenses are lowered. There is a lot of research going on here among data scientists and machine learning professionals. It is of the utmost importance to enhance both the dependability of fire alarms and the speed at which they are relayed. The system that is being presented is called Learning-based Remote Fire Detection (LBRFD), and it is designed to do the following steps: acquire information from computer vision, preprocess it, and then pool it using an area of suggestions. Object identification methods and computer vision are utilized to their full potential in order to get these results.

This is followed by the use of the region-based item identification approach, which makes use of the Region of Interest (RoI) to classify those thoughts as either fire or non-fire entities. The learning scheme that is recommended is called LBRFD, and it is a system that is the most advanced, lightning-fast, and pinpoint-accurate item recognition system. It is via the reuse of classifiers or localizers that earlier detection systems are able to locate items. In order to alter a photograph, they use the model to change it at various sizes and positions. For the purpose of analyzing the entire image, it makes use of a predictive learning strategy and adopts an entirely new approach, recognizing high-scoring sections as detections. Through the use of image segmentation, this network is able to forecast the boundary bounds of each region as well as the alternative outcomes. In comparison to systems that are based on classifiers, the weighting of these bounding boxes with the predicted probability offers significant advantages. The forecast is based on the total context of the image thanks to the fact that it takes into consideration the entire picture when testing is being done. Additionally, it is able to come up with predictions with only a single network assessment, whereas conventional systems require hundreds of evaluations for a single photo. Due to the fact that every cell in the grid can only have a single class and can only foresee two boundaries, limit area projections are heavily confined by the spatial limits imposed by LBRFD. Due to this spatial constraint, our model is only able to foresee a limited number of items that are directly near to it. When it comes to our strategy, it is difficult to deal with groups of little things, such as a cluster of trees. Due to the fact that this model is trained to determine boundary with limits based on data, it has trouble generalizing objects that have proportions of aspect or topologies that are either extremely unique or

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unconventional. For a clear image of the proposed system architecture, see the accompanying diagram (Fig. 2).

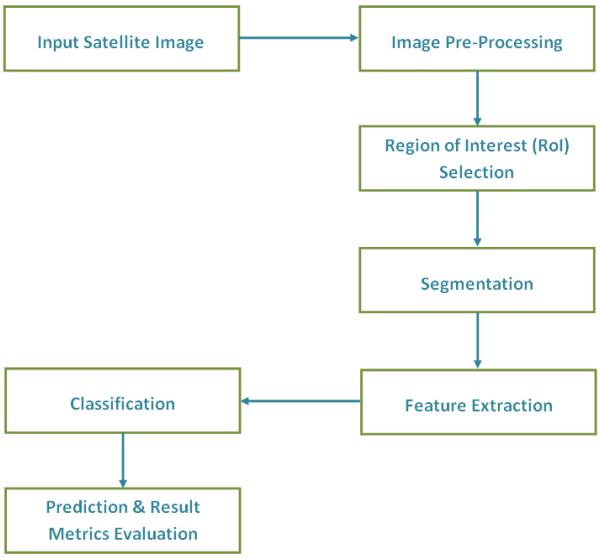

Fig.2 System Architecture

(i) Data Acquirement: The data is presented as video frames captured from surveillance cameras and other publicly available sources, such as Kaggle. However, for the sake of ease of use, specifically made videos will be used for testing and training purposes. Gathering these kinds of records using fire is a laborious process. The next step is to save both the fire-related and non-related frames. Subsequently, the dataset is partitioned into sets for training and testing. Careful attention to detail is required since skewed input data corrupts the entire network outputs, rendering the system inaccurate.

(ii) Pre-Processing: Image preparation follows model training for producing top-notch learning models. In this step, the data is processed and cleaned up so that it is ready for use. Eliminating background noise and other unwanted objects is an important part of data preparation. In order to avoid unpleasant outcomes, it is imperative that the proposed algorithm be supplied accurate information.

(iii) Extracting Features: A thorough understanding of fire's properties and its visual appearance is necessary for the proposed approach to make accurate fire detection. When we see fire, it's easy for our eyes to recognize it. Depending on the fuel it consumes, fire can take on a variety of forms and movements, emitting a reddish hue. In this investigation, fire and smoke are detected based on their form, colour, and velocity. We take the different frames from the training set and pull out their traits. Neural networks employ feature extraction networks to gather these characteristics for use in object identification, which is suggested approach powered method. After the characteristics are retrieved, the video frames are categorized as either being in a fire situation or not. Image characterizations and limits are used to extract the attributes.

(iv) Constructing the Model: In order to construct a model, the proposed system is fed the features that have been extracted and it can more precisely identify fires with the aid of this model, which is a collection of thresholds. By applying what it has learned from the features it has collected, the model can evaluate fresh data with more precision.

IV. Results and Discussions

Due to the obvious importance of reliability and system performance, it is crucial to validate the deep learning model. This validation step uses a separate collection of video footage from the set of data that was used to build the algorithm. The testing showed that the technique achieved an accuracy of around 98.33% with the set of validation results. The results of the suggested method are quite encouraging, and the system achieved a training accuracy rate of 98.54% in detecting fires. Learning based remote fire detection (LBRFD) showed promise in compared to prior Convolutional Neural Networks (CNNs) for fire detection. The system calculates and reduces false alarm rates by dynamically mixing multiple training data sets using a fully connected network. After that, the information is sent into an algorithm that can decide if a fire is present. Accuracy and metrics are top-notch generally, with the exception of a few photos where there were minor detecting issues. The biggest drawback is the little slowness it experiences due to the increased computing resources needed to provide results. The error rating can be reduced by improving the information purification process. Implementation should be done with a minimum rate of alarms that are inaccurate. The input image, output of the pre-processing and the feature extraction step of the suggested strategy are shown in Figures 3, 4 and 5, respectively.

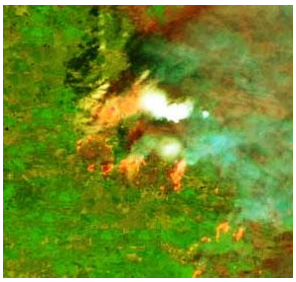
Fig.3 Input Image

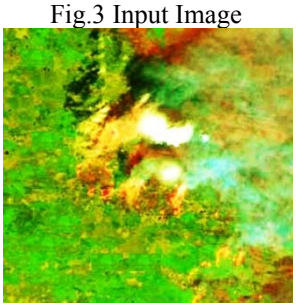
Fig.4 Pre-Processing

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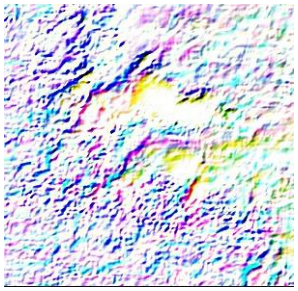

Fig.5 Feature Extraction

Figures 6, 7, and 8 show the results of the proposed scheme's segmentation and classification stages, as well as the Region of Interest (RoI) selection.

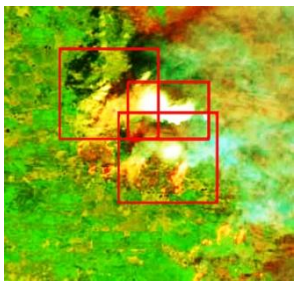

Fig.6 RoI Selection

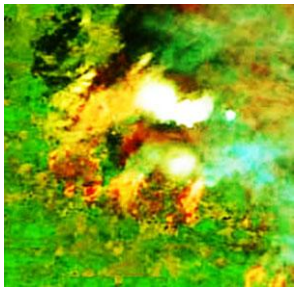

Fig.7 Segmentation

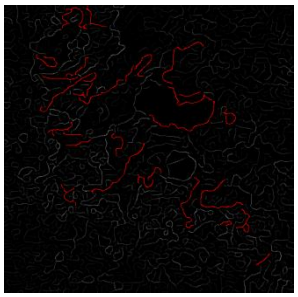

Fig.8 Classification

Figure 9 shows the results of a cross-validation test between the suggested LBRFD scheme and a traditional learning model, CNN, to see how well the former performs in making predictions. Table-1 provides a descriptive representation of the same.

Table-1: Accuracy Analysis

Epochs	CNN (%)	LBRFD (%)
100	92.46	98.67
125	91.34	98.45
150	90.67	97.16
175	92.59	97.92
200	91.72	98.09
225	89.63	96.19
250	88.45	96.72
275	92.67	97.35
300	90.28	97.83
325	90.08	98.45
350	89.88	98.67
375	92.65	98.83
400	91.39	97.91
425	89.47	98.54
450	89.07	98.24
475	88.87	98.29
500	88.67	98.33

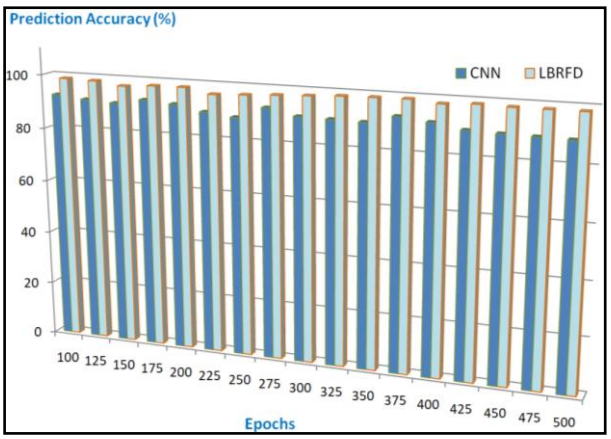

Fig.9 Accuracy Analysis

V. CONCLUSION

Here, we provide a method that uses LBRFD and CNN to identify forest fires in images. Images of flames are used for training and testing by these algorithms. We present the model that has been suggested and offer an adaptive pooling approach that uses colour characteristics to solve the problem of how typical pooling approaches in LBRFD may sometimes degrade picture features. In order to determine how different factors, such as learning rate and batch size, impact the LBRFD's performance, we do experiments. By doing these trials, we are able to ascertain the most suitable algorithm settings. Our method relies on colour cues to identify the potential flame region. This method improves characteristics like form and texture while decreasing the image characteristics of the buried layer's non-flame region. We keep picture data intact by using adaptive pooling. We evaluate the fragmented fire zone pictures' identification rates against those of the unsegmented originals. The outcomes prove that our suggested approach is practical for identifying forest fires in images and achieves a high rate of recognition. Our long-term goal is to refine LBRFD's

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forest picture identification using the updated pooling method. Through time-consuming comparisons with other algorithms, we intend to delve further into our investigation of recognition rates. We may evaluate our algorithm's performance and effectiveness in comparison to other methods by doing those comparisons.

REFERENCES

[1] H. Dang-Ngoc and H. Nguyen-Trung, "Aerial Forest Fire Surveillance - Evaluation of Forest Fire Detection Model using Aerial Videos", 2019 International Conference on Advanced Technologies for Communications (ATC), pp. 142-148, 2019.

[2] G. R et al, "A Survey On Sar Images Using Image Processing Techniques," (ICACCS), Coimbatore, India, 2020, pp. 1097-1100, doi: 10.1109/ICACCS48705.2020.9074261.

[3] Na Li, Jiameng Xue and Hongan Li, "'An Adaptive Detection Method for Early Smoke of Coal Mine Fire Based on Local Features'", 2022 International Conference on Image Processing and Media Computing (ICIPMC).

[4] S. R et al, "Comparison and analysis of Rice Blast disease identification in Greenhouse Controlled Environment and Field Environment using ML Algorithms," (ACCAI), Jan. 2022, Published, doi: 10.1109/accai53970.2022.9752538.

[5] T. Preeti, S. Kanakaraddi, A. Beelagi, S. Malagi and A. Sudi, "Forest Fire Prediction Using Machine Learning Techniques", 2021 International Conference on Intelligent Technologies (CONIT), pp. 1-6, 2021.

[6] Y. Zhang, Y. Zhang and Z. Yu, "A Solution for Searching and Monitoring Forest Fires Based on Multiple UAVs", 2019 International Conference on Unmanned Aircraft Systems (ICUAS), pp. 661-666, 2019.

[7] A. L. Latifah, A. Shabrina, I. N. Wahyuni and R. Sadikin, "Evaluation of Random Forest model for forest fire prediction based on climatology over Borneo", 2019 International Conference on Computer Control Informatics and its Applications (IC3INA), pp. 4-8, 2019.

[8] A. Jayakumar, A. Shaji and L. Nitha, "Wildfire forecast within the districts of Kerala using Fuzzy and ANFIS", 2020 Fourth International Conference on Computing Methodologies and Communication (ICCMC), pp. 666-669, 2020.

[9] Aditi Mukte, Urvashi Agrawal and Rahul Agrawal, "Smart Data Transfer for Data Monetization", IEEE International Conference on Computational Intelligence & Computing Applications 2021, Nov. 2021.

[10] Shixiao Wu and Libing Zhang, "'Using Popular Object Detection Methods for Real-Time Forest Fire Detection'", 2018 11th International Symposium on Computational Intelligence and Design (ISCID).

[11] B. Arteaga, M. Diaz and M. Jojoa, "Deep Learning Applied to Forest Fire Detection", 2020 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT), pp. 1-6, 2020.

[12] Y. Chen et al., "UAV Image-based Forest Fire Detection Approach Using Convolutional Neural Network", 2019 14th IEEE Conference on Indust rial Electronics and Applications (ICIEA), pp. 2118-2123, 2019.

[13] R. P. Sadewa, B. Irawan and C. Setianingsih, "Fire Detection Using Image Processing Techniques with Convolutional Neural Networks", 2019 International Seminar on Research of Information Technology and Intelligent Systems (ISRITI), pp. 290-295, 2019.

[14] G. D. Georgiev, G. Hristov, P. Zahariev and D. Kinaneva, "Forest Monitoring System for Early Fire Detection Based on Convolutional Neural Network and UAV imagery", 2020 28th National Conference with International Participation (TELECOM), pp. 57-60, 2020.

[15] F.M. AnimHossain, Youmin M.Zhang, and Masuda AkterTonima, "Forest fire flame and smoke detection from UAV-captured images using fire-specific color features and multi-color space local binary pattern", Journal of Unmanned Vehicle Systems, https://doi.org/10.1139/juvs-2020-0009, 2020.

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